# Neural Network Learning of Black-Scholes Equation for Option Pricing

## A presentation on De Souza Santos and Ferreira's (2024) paper
## MH4110 Partial Differential Equations

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# What is an Option?

- Derivatives are financial contracts which derive their value from an underlying asset
- Futures, Forwards, Swaps, Options
- Options:
  - Gives the right, not the obligation, to buy/sell asset at specified price on specified date
  - Call gives right to buy
  - Put gives right to sell
  - American style can be utilised at any time up until expiration date
  - European style can only be utilised on expiration date

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## Black-Scholes Background

- Developed in 1970s by Fischer Black, Myron Scholes, and Robert Merton
- Provides an analytic solution to a second order parabolic partial differential equation by a transformation to the heat equation
- Leads to conclusion that options pricing dynamic is similar to heat dissipation
- Based on European style options
- American style options pricing can only be done numerically, not analytically

## Black-Scholes Equation

The Black-Scholes equation is a second-order parabolic PDE. For a stock which does not pay dividends, we have:

$$\frac{\partial c}{\partial t} + rS\frac{\partial c}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 c}{\partial S^2} = rc$$

- $t$ = time
- $c$ = call option price
- $p$ = put option price
- $S$ = spot stock price
- $r$ = risk free interest rate
- $\sigma$ = stock volatility
- $S_0$ = spot stock price at time zero
- $K$ = strike price
- $T$ = time to maturity

Solutions to this PDE are the formulas for European call and put options prices.

## Boundary Conditions

- Boundary conditions come from the payoff, given by:

$$\text{payoff} = \text{strike} - \text{spot}$$

- For a European call option, we have the terminal condition:

$$\text{payoff} = \max\{S_T - K, 0\}$$

- For a European put option, we have the terminal condition:

$$\text{payoff} = \max\{K - S_T, 0\}$$

where $t = T$ because this is the exact moment of expiration of the option

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## Transformation to the Heat Equation

Firstly, we define:

$$x = \ln\left(\frac{S(t)}{K}\right) \qquad \text{and} \qquad \tau = \frac{1}{2}\sigma^2(T-t)$$

Take log of price as stock price follows geometric Brownian motion.
Define rescale of time as time scaled by volatility.
Define mapping from $c(s,t)$ to $f(x,\tau)$ with strike price $K$ as scale factor:

$$c(s,t) = Kf(x,\tau)$$

Substituting in the partial derivatives from Black-Scholes allows us to find the PDE:

$$\frac{\partial f}{\partial \tau} = \frac{\partial^2 f}{\partial x^2} + \left(\frac{2r}{\sigma^2} - 1\right)\frac{\partial f}{\partial x} - \frac{2r}{\sigma^2} f$$

which can be re-written as:

$$\frac{\partial f}{\partial \tau} = \frac{\partial^2 f}{\partial x^2} + (k-1)\frac{\partial f}{\partial x} - kf$$

where $k = \frac{2r}{\sigma^2}$. We see this is a PDE with similar form to the heat equation, but with extra terms $(k-1)\frac{\partial f}{\partial x}$ and $-kf$.

To make the extra terms disappear, we define a new mapping, where the exponential removes drift:

$$f(x,\tau) = \exp(\alpha x + \beta\tau)\, u(x,\tau)$$

Drift is the expected return of an asset over time. We remove it and replace it with the risk-free rate to create a risk-neutral environment. The exponential function removes drift. We then solve and choose $\alpha$ and $\beta$ such that the extra terms are removed, so we find:

$$\alpha = -\tfrac{1}{2}(k-1) \qquad \text{and} \qquad \beta = -\tfrac{1}{4}(k+1)^2$$

which leads to a direct relationship between actual price of the option $c(s,t)$, and the modified price $u(x,\tau)$. Substituting $\alpha$ and $\beta$ we find this relationship is given by:

$$c(s,t) = K\, u(x,\tau)\, \exp\!\left(-\tfrac{1}{2}(k-1)\, x - \tfrac{1}{4}(k+1)^2\, \tau\right)$$

which will later be used by the ANN.

After substituting this relationship into the original Black-Scholes equation, we are left with the heat equation. So we see that the Black-Scholes equation can be transformed to the heat equation, where $x$ is the price of the stock after transformations, $\tau$ is the unit of time after transformations, and $u$ is the price of the option after transformations:

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}, \qquad -\infty < x < \infty, \qquad \tau > 0$$

Where we have the payoff boundary condition:

$$u(x,0) = \max\left\{\exp\left(\tfrac{1}{2}(k+1)\,x\right) - \exp\left(\tfrac{1}{2}(k-1)\,x\right),\ 0\right\}$$

## Final Solution

We then use the Gaussian kernel and substitute the solution $u(x,\tau)$ of this heat equation into our expression:

$$c(s,t) = K\,u(x,\tau)\,\exp\left(-\tfrac{1}{2}(k-1)\,x - \tfrac{1}{4}(k+1)^2\,\tau\right)$$

Through the method we have learnt in class, and lots of cancellations, we can obtain the solutions to the Black-Scholes equation, which shows the pricing of European call and put options:

$$c = S_0 N(d_1) - K\exp(-rT)\,N(d_2) \qquad p = K\exp(-rT)\,N(-d_2) - S_0 N(-d_1)$$

where

$$d_1 = \frac{\ln\left(\dfrac{S_0}{K}\right) + \left(r + \dfrac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}} \qquad \text{and} \qquad d_2 = d_1 - \sigma\sqrt{T-t}$$

$d_1$ and $d_2$ are parameters which represent risk-adjusted probabilities within the normal CDF $N(x)$, which comes from the Gaussian integral.

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# The Analytical Solution and Its Limits

**The Black-Scholes-Merton formula** [1] gives a closed-form option price:

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

**Assumptions required:**

- Constant volatility $\sigma$
- Log-normal returns
- No dividends
- Frictionless markets

## Market Reality

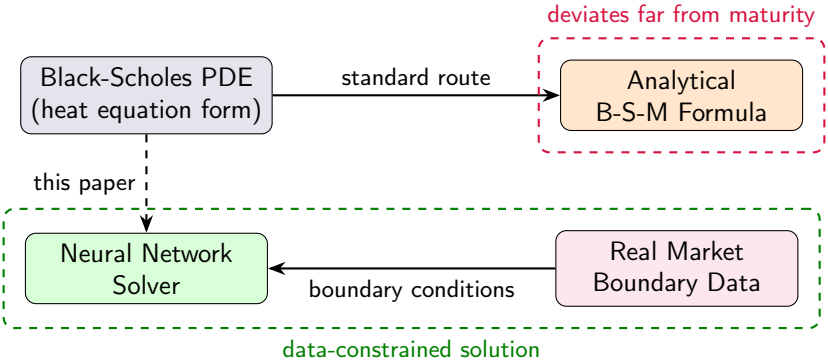
For dates **far from maturity**, the analytical solution tracks the *stock price*, not the *option price*.

It only converges to market prices near expiration.

## Key Claim [3]

The issue is not the PDE itself. The real-world divergence comes from the **analytical solution** applied in practice.

# PDE-Constrained Learning: The Core Idea




## This is not regression

The network does not curve-fit option prices directly. It must learn a function that **satisfies the PDE** as a constraint while matching real market boundary data.

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# (A) The PDE as an Optimization Problem

**Standard PDE formulation:**

$$D(u) - F = 0$$

**Network output** $\hat{u}$: a candidate solution.

**Residual:**

$$\mathcal{R}(\hat{u}) = D(\hat{u}) - F$$

**Training objective:**

$$\min_{\theta} \|\mathcal{R}(\hat{u}_\theta)\|^2$$

When $\mathcal{R} = 0$ everywhere, $\hat{u}$ is an exact solution.

## Physics-Informed Neural Networks

This is the core of **PINNs**, first proposed by Lagaris, Likas, and Fotiadis [7] and extended by Raissi, Perdikaris, and Karniadakis [10].

The PDE residual acts as a **physics-based regularizer** constraining the solution space.

For Black-Scholes: $F = 0$ (homogeneous heat equation), $D(\cdot) = \partial_\tau - \partial_{xx}$

## (B) Enforcing Boundary Conditions

**Problem:** Minimizing the residual alone does not guarantee the network satisfies the initial condition (option payoff at expiration).

**Solution: modified trial function [7, 2]:**

$$\tilde{u}(x,\tau) = \underbrace{u_{\tau_0}(x)}_{\text{known payoff}} + \underbrace{\left[1 - e^{-(\tau-\tau_0)}\right]}_{\substack{=0 \text{ at } \tau=\tau_0 \\ \to 1 \text{ as } \tau \text{ grows}}} \cdot \hat{u}(x,\tau)$$

### Effect

At $\tau = \tau_0$: $\tilde{u} = u_{\tau_0}(x)$ **exactly**

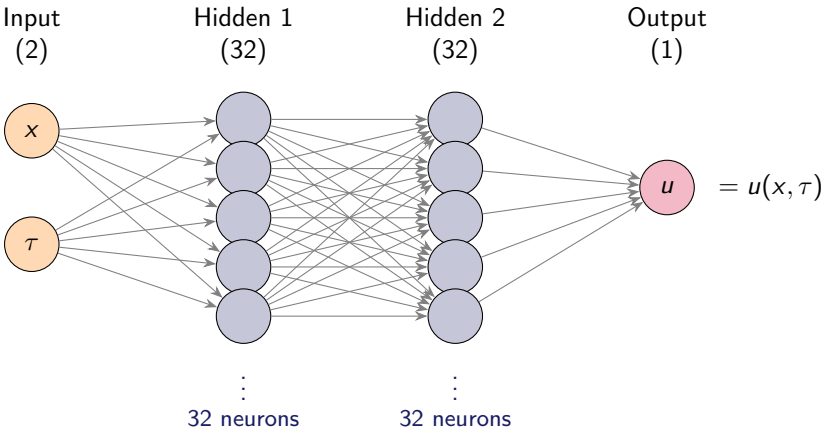
As $\tau$ grows: $\tilde{u} \approx \hat{u}(x,\tau)$

### Why not a penalty term?

Penalty-based enforcement creates competing gradients. The modified trial function bakes the constraint into the architecture: satisfied *analytically*, not approximately.

General form: $\tilde{u}(x,\tau) = A(x,\tau;\, x_{\text{bdy}}, \tau_0)\, \hat{u}(x,\tau)$, where $A$ encodes all boundary constraints.
Implemented via **NeuroDiffEq** [2].

# (C) Network Architecture: Structure




MLP with architecture $2 \to 32 \to 32 \to 1$ [3].
Inputs are transformed coordinates $(x, \tau)$; output is the generalised option price.

# (C) Network Architecture: Training Details

| Component | Choice |
|---|---|
| Architecture | MLP $(2 \to 32 \to 32 \to 1)$ |
| Input $x$ | $\ln(S/c)$, scaled by $K$ |
| Input $\tau$ | $\frac{1}{2}\sigma^2(T-t)$ |
| Activation | tanh |
| Optimizer | Adam [6] |
| Epochs | 30,000 |
| Framework | PyTorch [9] |
| Library | NeuroDiffEq [2] |

## Why tanh? [7, 5]

Smooth and differentiable everywhere. Computing the PDE residual requires differentiating the network output w.r.t. inputs, so the activation must be differentiable across the whole domain.

## Why Adam? [6]

Adapts the learning rate per parameter using gradient moment estimates. Converges faster than vanilla SGD, especially when gradient scales vary across the network.

# Coordinate Transformation: Connecting $u$ to Real Prices

The network operates in transformed coordinates [3]. Recovering the real option price requires two substitutions:

## Variable Substitutions (Black-Scholes $\rightarrow$ Heat Equation)

$$x = \ln\left(\frac{S_t}{K}\right), \qquad \tau = \tfrac{1}{2}\sigma^2(T-t)$$

$$c(S,t) = K\,\hat{u}(x,\tau)\,\exp\left(-\tfrac{1}{2}(k-1)x - \tfrac{1}{4}(k+1)^2\tau\right), \quad k = \frac{2r}{\sigma^2}$$

**Inputs to the network:**

- $x$: log-scaled, normalised price ratio
- $\tau$: variance-scaled time to maturity

**Output from the network:**

- $u(x,\tau)$: generalised (transformed) price
- Back-transform gives $c(S,t)$ in BRL

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# Experimental Setup: Data

**Market:** Brazilian Stock Exchange (B3) [3]

| Stock | Ticker | Series |
|---|---|---|
| Petrobras | PETR4 | PETRA (12 options) |
| | | PETRD (11 options) |
| Vale | VALE3 | VALED (10 options) |

- Series A: maturity **January 2023**
- Series D: maturity **April 2023**
- **Call options only** (puts excluded due to low volume)
- Interest rate: SELIC = **13.75%** p.a. (fixed)

## Why Petrobras and Vale?

Highest trading volume on B3, giving dense and reliable time series. Other stocks such as Banco do Brasil and WEG had too many days without trades to be usable.

## Data transformation

Raw market prices are converted to $(x, \tau)$ coordinates before training. The payoff function serves as the initial boundary condition in transformed space.

# Error Metrics [4]

| Metric | Formula | What it measures |
|---|---|---|
| **MAE** | $\frac{1}{N}\sum_{i=1}^{N}\left\lvert Y_i - \hat{Y}_i\right\rvert$ | Average absolute error in BRL. Lower is better. |
| **MSE** | $\frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2$ | Squares errors; penalises large misses more heavily. |
| **MAPE** | $\frac{1}{N}\sum_{i=1}^{N}\left\lvert\frac{Y_i - \hat{Y}_i}{Y_i}\right\rvert$ | Relative error; comparable across different price levels. |
| **POCID** | $100 \cdot \frac{\sum_{i=1}^{N} D_i}{N}, \quad D_i = 1_{[(Y_i - Y_{i-1})(\hat{Y}_i - \hat{Y}_{i-1}) > 0]}$ | Fraction of days the model correctly predicted price direction. |
| **ARV** | $\frac{\sum_{i=1}^{N}(\hat{Y}_i - Y_i)^2}{\sum_{i=1}^{N}(\hat{Y}_i - \bar{Y})^2}$ | Below 1 beats a naive mean predictor; 0 is a perfect model. |

# What to Look For in the Results

**Graph color key [3]:**

- **Blue** — actual option market price
- **Purple** — underlying stock price (SPOT)
- **Magenta** — analytical B-S solution (BLS)
- **Green** — neural network prediction (NN)

**What to look for:** Green tracks blue throughout. Magenta follows purple and only converges to blue near expiration.

## Benchmark [8]

Hong Kong options forecasting with NNs and SVR.

Best MAE reported: **14.3**

This paper (Petrobras): MAE $\approx$ **2.6**
This paper (Vale): avg. MAE $\approx$ **5.6**

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# Results

The experimental results demonstrate that the Multi-Layer Perceptron (MLP) neural network successfully learned to solve the modified Black-Scholes equation, providing more accurate option pricing forecasts than traditional analytical solutions.

# Table 1

TABLE 1. Statistical Errors for MLP Neural Network Modeling the Petrobras Options with Black-Scholes Model - A Series for PETR4. The best value for each error measure is in boldface.

| PETRA | K (BRL) | MAE | MSE | MAPE | POCID | ARV | N |
|---|---|---|---|---|---|---|---|
| **332** | 24.76 | **0.980** | **1.295** | 1.257 | 75.000 | 0.004 | 61 |
| **357** | 26.26 | 1.145 | 1.641 | 2.590 | 77.358 | 0.023 | 54 |
| **356** | 25.76 | 1.345 | 2.064 | 2.733 | 82.353 | 0.020 | 52 |
| **366** | 26.76 | 1.141 | 2.337 | 2.114 | 71.667 | 0.020 | 61 |
| **342** | 25.26 | 1.530 | 2.706 | 2.357 | **84.314** | 0.019 | 52 |
| **266** | 17.76 | 5.749 | 37.778 | 0.800 | 53.333 | 0.013 | 61 |
| **391** | 22.76 | 1.179 | 1.888 | **0.528** | 78.333 | **0.003** | 61 |
| **321** | 22.26 | 1.670 | 4.563 | 0.533 | 71.667 | 0.006 | 61 |
| **282** | 18.76 | 6.231 | 42.81 | 1.084 | 72.340 | 0.019 | 48 |
| **53** | 15.26 | 2.800 | 10.538 | 0.598 | 46.154 | 0.023 | 40 |
| **108** | 13.76 | 8.418 | 73.457 | 0.820 | 72.414 | 0.032 | 30 |
| **362** | 27.26 | 1.011 | 1.942 | 2.163 | 68.333 | 0.020 | 61 |

# Figure 1

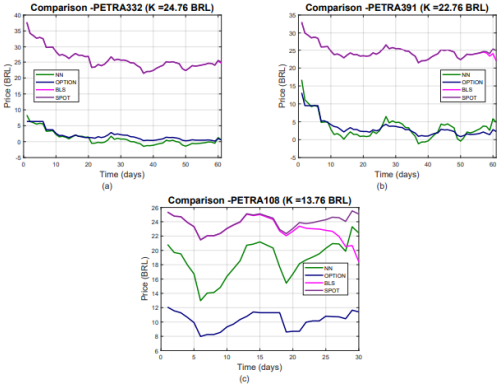


FIGURE 1. (a) Results for PETRA332 (MSE minor) (b) Results for PETRA391 and (c) results for PETRA108 (MSE major). (a) and (b) are the best NN prediction cases, and (c) is the worst NN prediction case. For all graphics, there are four curves: the price of the underlying PETR4 stock - OPTION, the blue curve; the price calculated by the Black-Scholes analytical solution - BLS, the magenta curve; the price computed by the ANN, the green curve; and, the option market price - SPOT, the purple curve.

# Table 2

TABLE 2. Statistical Errors for MLP Artificial Neural Network (ANN) Modeling the Petrobras Options with Black-Scholes Model - D Serie for PETR4. The best value for each error measure is in boldface.

| PETRD | K (BRL) | MAE | MSE | MAPE | POCID | ARV | N |
|---|---|---|---|---|---|---|---|
| **291** | 28.26 | 0.727 | 0.906 | 1.922 | 16.667 | 0.028 | 61 |
| **222** | 26.76 | 1.114 | 1.664 | 3.324 | 8.333 | 0.069 | 61 |
| **266** | 27.76 | **0.670** | **0.751** | 2.115 | 11.667 | 0.054 | 61 |
| **271** | 27.26 | 0.806 | 1.020 | 2.507 | 13.333 | 0.067 | 61 |
| **220** | 26.26 | 1.198 | 1.997 | 2.231 | 10.000 | 0.056 | 61 |
| **227** | 25.26 | 1.539 | 3.602 | 0.938 | 10.000 | 0.030 | 61 |
| **202** | 25.76 | 1.301 | 2.484 | 1.377 | 6.667 | 0.040 | 61 |
| **194** | 23.76 | 2.173 | 6.636 | **0.771** | 16.667 | 0.024 | 61 |
| **183** | 24.76 | 1.798 | 4.599 | 1.089 | 11.667 | 0.036 | 61 |
| **198** | 22.76 | 2.848 | 9.541 | 0.855 | 33.333 | 0.019 | 61 |
| **122** | 22.26 | 2.751 | 8.501 | 0.822 | **35.000** | **0.018** | 61 |

# Figure 2

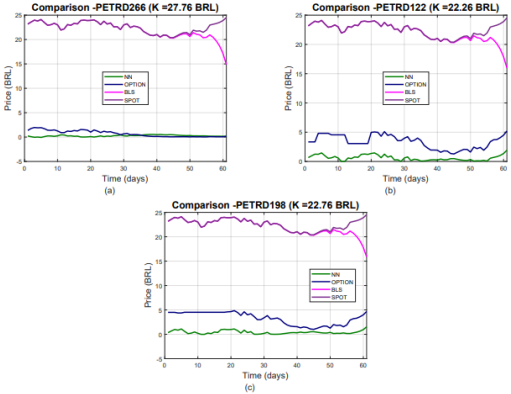


FIGURE 2. (a) Results for PETRD266 (MSE minor), (b) Results for PETRD122 and (c) results for PETRA198 (MSE major). (a) and (b) are the best NN prediction cases, and (c) is the worst NN prediction case. For all graphics, there are four curves: the price of the underlying PETR4 stock - OPTION, the blue curve; the price calculated by the Black-Scholes analytical solution - BLS, the magenta curve; the price computed by the ANN, the green curve; and, the option market price - SPOT, the purple curve.

# Table 3: Vale Options (VALED)

TABLE 3. Statistical Errors for MLP Neural Network Modeling the Vale Options with Black-Scholes Model - D Series for VALE3. The best value for each error measure is in boldface.

| VALED | K (BRL) | MAE | MSE | MAPE | POCID | ARV | N |
|---|---|---|---|---|---|---|---|
| **80** | 78.97 | 5.164 | 32.565 | 0.792 | 58.333 | **0.019** | 61 |
| **77** | 75.97 | 4.539 | 26.032 | 0.547 | 77.778 | 0.025 | 46 |
| **765** | 76.97 | **3.435** | **15.172** | 0.540 | **78.378** | 0.029 | 38 |
| **79** | 77.97 | 6.727 | 59.320 | 0.771 | 60.000 | 0.021 | 61 |
| **743** | 74.97 | 4.213 | 22.747 | 0.510 | 71.429 | 0.033 | 36 |
| **81** | 79.97 | 5.617 | 41.571 | 0.850 | 56.667 | 0.021 | 61 |
| **83** | 81.97 | 4.901 | 35.771 | 0.951 | 73.214 | 0.025 | 57 |
| **82** | 80.97 | 5.536 | 42.683 | 0.887 | 66.667 | 0.022 | 61 |
| **75** | 73.97 | 4.814 | 29.249 | **0.484** | 60.976 | 0.028 | 42 |
| **655** | 66.97 | 10.604 | 115.290 | 0.715 | 20.690 | 0.034 | 30 |

- Errors vary significantly across contracts
- In-the-money options (lower strike) generally show lower errors
- Higher errors tend to correspond to less frequently traded contracts
- **Key insight:** NN performance depends on the option's trading activity and moneyness

# Figure 3: Model Comparison (VALED)

NN LEARNING OF BLACK-SCHOLES EQUATION FOR OPTION PRICING 13

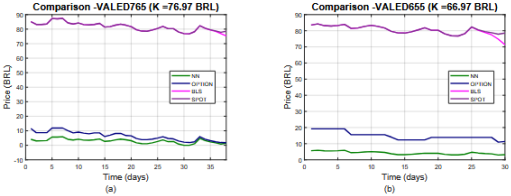


FIGURE 3. (a) Results for VALED765 (MSE minor, the bets NN prediction) and (b) results for VALED655 (MSE major, the worse NN prediction). For both graphics, there are four curves: the price of the underlying PETR4 stock - OPTION, the blue curve; the price calculated by the Black-Scholes analytical solution - BLS, the magenta curve; the price computed by the ANN, the green curve; and, the option market price - SPOT, the purple curve.

**Each plot shows three curves:**

- Market price (OPTION)
- Black-Scholes solution (BLS)
- Neural network output (NN)

- (a) Best NN case: low MSE, NN tracks market closely
- (b) Worst NN case: high MSE, NN deviates more noticeably

**Key insight:** NN outperforms BLS overall, but accuracy is not uniform across all contracts

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# Evaluation and Conclusion

So far, we have seen how the Black-Scholes equation can be derived from financial principles and transformed into a PDE, and how a neural network can be trained to approximate its solution.

In this final section, we will evaluate this approach by briefly discussing its strengths, limitations, and implications for financial modelling.

## Strengths of the Neural Network Approach

Recall the Black-Scholes equation:

$$\frac{\partial V(S,t)}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V(S,t)}{\partial S^2} + rS\frac{\partial V(S,t)}{\partial S} - rV(S,t) = 0$$

We want to approximate the option value $V(S,t)$. By utilising neural networks, we have the following advantages:

- Handles higher dimensions
- Applicable Without Closed-Form Solutions

## Advantage 1: Handles Higher-Dimensional Problems

In more realistic financial models, the option price may depend on more than one variable.

For example, an option depending on two assets can be written as

$$V = V(S_1, S_2, t)$$

This leads to a larger PDE involving derivatives with respect to both asset prices.

As the number of variables increases, solving the PDE becomes more complicated. Neural networks can approximate these higher-dimensional solutions more easily.

## Advantage 2: Applicable Even Without Closed-Form Solution

Not all options have a simple formula like the standard European call.

For example, an option depending on the average stock price over time:

$$V = V\left(S(t), \frac{1}{T}\int_0^T S(t)dt, t\right)$$

This makes it difficult to obtain a closed-form solution.

However, the Black–Scholes PDE still applies as it gives a rule that the option price must satisfy, even for these complex cases.

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## Limitation 1: High Computational Cost

- Neural networks must be trained using many sampled points from the PDE domain
- Each training step requires computing derivatives of the network output (PDE residual)
- This makes training significantly slower than standard finite difference methods for simple PDEs
- For the Black–Scholes model, closed-form or fast numerical methods are often more efficient

## Limitation 2: Harder to Interpret

- Neural networks are often described as black-box models because the internal decision process is not easily interpretable
- Unlike analytical Black–Scholes formulas, it is difficult to see how individual inputs (e.g. volatility or strike price) affect the output
- Model accuracy depends strongly on training data and network design choices
- This lack of transparency can be a concern in financial settings where interpretability is important

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## Implications for Financial Modelling

- Neural networks provide a flexible framework for solving option pricing PDEs
- They can incorporate market data, potentially capturing effects not reflected in classical models
- Useful for high-dimensional or complex derivatives where analytical formulas are unavailable
- Even when closed-form solutions exist, they may not fully match observed market prices

# Conclusion

- The Black–Scholes equation provides a fundamental PDE framework for option pricing in financial mathematics
- While classical methods rely on analytical solutions or grid-based numerical schemes, these approaches can be limited in complexity and scalability
- Neural networks offer an alternative approach by directly approximating the PDE solution in a flexible way
- This makes them particularly useful for high-dimensional problems and financial models
- However, issues such as computational cost, interpretability, and model validation remain important in practical applications

[1] Fischer Black and Myron Scholes. "The Pricing of Options and Corporate Liabilities". In: *Journal of Political Economy* 81.3 (1973), pp. 637–654.

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